The Wow! Signal Architecture: Filtering Stochastic LLM Hallucinations via Venturi Compression and L0-ISM Deterministic SMT Solvers

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Abstract

The integration of Large Language Models (LLMs) into critical infrastructure (e.g., data center cooling, power grids) presents a severe risk due to stochastic drifting and ungrounded generation, commonly known as hallucinations. Drawing inspiration from the astrophysical "Wow! Signal" of 1977—a discrete, time-bound, narrow-band anomaly successfully isolated from the high-entropy cosmic background—this paper proposes the Invariant State Matrix (L0-ISM) architecture. By conceptualizing a "Venturi compression nozzle" to quantize multi-dimensional neural outputs into a 19-state discrete vector $[-9, +9]$, and applying strict cyber-physical invariants (e.g., thermodynamic budget $P + T = Q$) via Satisfiability Modulo Theories (SMT) solvers like Z3, we achieve real-time, deterministic filtration of AI hallucinations. Failed verification triggers an immediate hardware-level -999 HARD STOP, ensuring zero latency propagation of physical anomalies.

1 Introduction: The High-Entropy Dilemma

In cognitive and computer sciences, generative models operating in the L7 (application) layer rely on frozen probability weights, making them inherently stochastic. When an LLM dictates control vectors for micro-Channel Cooling Distribution Units (mCDU) or DC power buses, its lack of real-time grounding inevitably yields non-physical instructions.

In astrophysics, the detection of the "Wow! Signal" demonstrated that intelligent, structured data can be isolated from infinite stochastic noise (the cosmic background) provided the receptor is hardware-locked to a universal invariant—in that case, the 1420 MHz neutral hydrogen emission line, bounded by a strict 72-second window. We posit that LLM governance requires a digital equivalent of this radio telescope: a deterministic hardware filter bounded by invariant physical laws.

2 Mathematical Apparatus: Venturi Compression

Unconstrained LLM output spaces contain billions of parameters. To prevent stochastic instructions from reaching physical actuators, the continuous probability output O_{LLM} must be forcibly compressed. We model this as a Venturi Nozzle transition, transforming turbulent, high-dimensional data into laminar, discrete vectors.

Let the output of the LLM be a continuous state matrix $\mathbf{M} \in \mathbb{R}^{N \times M}$. The Venturi function Φ_v acts as a quantization operator, projecting $\mathbf{M}$ onto the discrete L0-Invariant State Matrix (L0-ISM) space $\mathcal{S}$, where $\mathcal{S} = \{-9, -8, \ldots, 0, \ldots, +8, +9\}$.

$$s_i = \Phi_v(O_{LLM}) = \text{sgn}(\omega \cdot O_{LLM}) \lfloor |\omega \cdot O_{LLM}| \rceil \quad \text{subject to} \quad s_i \in [-9, 9] \tag{1}$$

Any output that cannot be compressed into this exact 19-state integer vector is natively rejected.

3 Data Physics and Classical Invariants

Once quantized, the proposed state vector s_i must be evaluated against physical reality. We utilize classical thermodynamics and control theory as absolute invariants.

3.1 Thermodynamic Invariant

For a given data center unit, the total heat dissipated $Q(t)$ must equal the sum of computing power $P(t)$ and ambient thermal load $T(t)$, bounded by the mCDU cooling capacity Q_{max}.

$$Q(t) = P(t) + T(t) \leq Q_{max} \tag{2}$$

If s_i necessitates a state change (e.g., increased clock speeds) that violates $\int (P(t)+T(t))dt \leq Q_{max}$, the condition mathematically fails.

3.2 Time-Bound Protocol (T-OTP)

Drawing from the 72-second lifecycle of the Wow! Signal, valid control vectors must arrive and execute within a strict temporal window Δt.

$$\forall s_i : \tau_{receive} - \tau_{issue} \leq \Delta t_{critical} \tag{3}$$

4 Z3 SMT Solver as the Hardware Watchdog

The quantized vector s_i and the invariant constraints are fed into a formal Z3 SMT solver. The solver tests the boolean satisfiability of the proposed state change against the unyielding laws of physics (Data Physics).

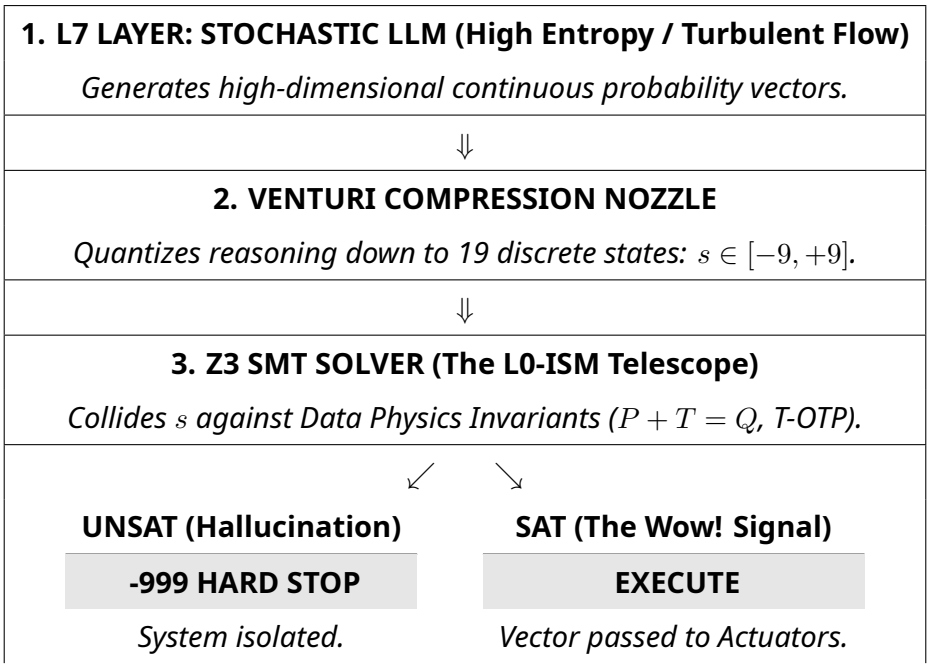
$$\mathcal{V}(s_i, \mathcal{I}_{physics}, \mathcal{I}_{time}) \models \begin{cases} \text{SAT} & \text{if conditions hold} \\ \text{UNSAT} & \text{if constraints violated} \end{cases} \tag{4}$$

If $\mathcal{V}$ evaluates to SAT, the vector is grounded and executes as a true, filtered "Wow! Signal." If it evaluates to UNSAT (i.e., the LLM hallucinates a physically impossible scenario), the system generates a hardware-level safety interrupt:

$$\text{If } \mathcal{V} = \text{UNSAT} \implies s_{out} = -999 \text{ (HARD STOP)} \tag{5}$$

5 System Architecture Diagram

Below is the formal block diagram of the proposed cascade validation process.



6 Conclusion

By establishing a deterministic neuro-symbolic framework, we prove that LLM hallucinations need not be corrected within the model itself. By treating them as stochastic noise and deploying a strict physical invariant structure (L0-ISM) paired with SMT verification, we guarantee that only physically viable ”Wow! Signals” reach critical hardware actuators, effectively eliminating fatigue, cavitation, and thermal degradation caused by non-deterministic software logic.